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User-Interactive Federated Learning for Automated Endpoint Intrusion Detection with Real-Time Visualization and Incremental Threat Adaptation

Grefith Gohel^{a,*}, Dev Jani^a, Vishal Barot^b

^a*L.D.R.P-ITR, Gandhinagar 382016, INDIA*

^b*GTU Institute of Technology and Research, Mehsana 384460, INDIA*

Abstract

The rapid proliferation of Internet of Things devices has intensified requirements for robust intrusion detection systems capable of operating within distributed, privacy-sensitive environments. Traditional centralized intrusion detection systems encounter significant challenges, including privacy vulnerabilities, communication latency, and limited adaptability to evolving threat landscapes. This paper presents a user-interactive federated learning-based intrusion detection system for automated endpoint monitoring in IoT networks. The proposed system employs Tshark for real-time packet capture and implements a hybrid Convolutional Neural Network-Long Short-Term Memory architecture using PyTorch for local threat detection. A blockchain-based federated learning framework ensures privacy-preserving collaborative model training across distributed IoT endpoints. The system incorporates a novel semi-supervised feedback mechanism through an intuitive Tkinter-based graphical interface, enabling non-technical users to refine model predictions incrementally. Privacy compliance is maintained through ephemeral data processing, where captured network traffic undergoes immediate analysis and deletion without persistent storage. Comprehensive evaluation across multiple datasets MQTT, UNSW-NB15, ToN-IoT, and CIC-IDS-2018 demonstrates robust generalization capabilities. Experimental results reveal initial detection accuracy of 99.66% with F1-score of 98.75% for malicious traffic classification. User feedback integration improves performance metrics by 8-18% across all evaluation criteria. The system maintains computational efficiency suitable for resource-constrained IoT environments while providing scalable security solutions for dynamic network conditions.

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* Corresponding author. Tel.: +91 8000712252.

E-mail address: grefithgohel90@gmail.com

1. Introduction

1.1. Background and Motivation

The proliferation of Internet of Things devices has fundamentally transformed connectivity paradigms across diverse sectors, simultaneously creating extensive attack surfaces that necessitate sophisticated intrusion detection systems capable of operating in distributed environments [1, 2]. Conventional centralized intrusion detection system frameworks exhibit inherent privacy limitations and communication latency issues, rendering them inadequate for dynamic IoT infrastructures [3]. Federated Learning has emerged as a promising paradigm for intrusion detection in distributed and privacy-sensitive environments, addressing fundamental limitations of centralized approaches including data confidentiality concerns, network communication overhead, and system scalability constraints [16, 22]. Recent advances in federated learning for IoT security have demonstrated significant improvements in privacy preservation and detection accuracy [23, 24]. Contemporary research emphasizes the integration of blockchain technology with federated learning frameworks to enhance security guarantees and prevent adversarial attacks [29, 32]. However, systematic reviews indicate persistent gaps in user-interactive mechanisms and real-time adaptability requirements [24]. Current research predominantly emphasizes algorithmic improvements while neglecting user-centric functionalities essential for practical deployment in non-expert environments [6, 7]. The expanding threat landscape, characterized by sophisticated attack vectors including advanced persistent threats and polymorphic malware, demands adaptive defense mechanisms capable of real-time threat mitigation and continuous model refinement [8, 9]. The integration of FL approaches with participatory user interaction bridges the gap between technological progress and operational feasibility, facilitating stakeholder involvement in system improvement while preserving holistic IoT protection [10]. This approach becomes even more significant looking ahead to the deployment of mission-critical infrastructure-based IoT where security violation has grave outcomes [11]. Inflexible classical detector systems cannot compete against sophisticated attack paradigms, while FL's locally distributed architecture allows site-specific customizations, reducing response time and integrating human expertise to stand out in threat identification capability [12, 13, 14].

1.2. Problem Statement

Traditional intrusion detection system deployments within Internet of Things devices face several challenges that undermine modern cybersecurity practices. Centralized data collection introduces serious privacy risks by aggregating sensitive information into unified points of vulnerability, which may expose user privacy to orchestrated breaches [15]. These systems also generate substantial latency through constant data transfers, posing particular difficulties for devices with restricted bandwidth and computational resources [18]. Moreover, static models fall short against dynamic attack patterns, such as advanced persistent threats and ransomware, leading to higher false negative rates and protracted threat responses [13]. Few easy intuitive interfaces make it challenging for non-technical users to install the systems, restricting their deployments within homes as well as small businesses [14]. The non-independent and identically distributed data characteristics across IoT devices complicate model training, potentially degrading federated learning performance [6]. Geographical and usage pattern variations generate statistically distinct data properties, challenging centralized model generalization [12]. Current FL-based IDS utilizing convolutional neural networks demonstrate potential but lack real-time visualization and feedback mechanisms essential for immediate endpoint responses [14]. These vulnerabilities facilitate cascading network failures, as demonstrated by compromised smart devices acting as conduits for wider attacks [13]. Regulatory frameworks, such as the General Data Protection Regulation, underscore the significance of data decentralization, thereby endorsing federated learning-based solutions [19]. This research is thus motivated to develop an automated, privacy-preserving intrusion detection system that enables continuous monitoring and progressive learning through user engagement [15].

1.3. Objectives of the Proposed Work

The primary objectives of this study involve developing an intrusion detection system that integrates federated learning, incorporating components that enable users to collaboratively and automatically identify threats on Internet of Things devices [1, 2]. This entails designing a comprehensive system architecture that combines local deep learning techniques with secure FL, ensuring optimal interaction among various components while preserving data

confidentiality [2]. The framework prioritizes the provision of real-time packet capture and visualization capabilities, offering user-friendly monitoring tools that empower non-expert users to effectively manage security scenarios [3]. Furthermore, the system introduces semi-supervised progressive learning mechanisms, facilitating adaptation to evolving concepts and attack strategies [4]. The proposed IDS is evaluated across multiple independent datasets and active network traffic to enhance generalizability, precision, recall, and adaptability compared to existing approaches [5]. This adaptive, user-centric security framework addresses current IDS limitations through continuous monitoring, ensuring timely threat detection while incorporating user input to refine the model, thereby improving resilience against escalating threats [6]. Comparative assessments across diverse IoT contexts, ranging from domestic to industrial applications, demonstrate its superiority over traditional and other FL-based methodologies [7]. This research delivers practical, scalable solutions that align technological advancements with usability to meet the dynamic security demands of IoT environments [8].

1.4. Contributions and Scope

The primary contributions of this study include the development of a novel user-participatory federated learning framework, underpinned by blockchain-assisted aggregation, to enhance security and confidentiality, thereby establishing a robust foundation for distributed intrusion detection systems (IDS) within Internet of Things infrastructures [9]. This framework introduces an automated endpoint IDS, leveraging PyTorch for local processing and Tshark for real-time traffic acquisition, enabling efficient edge-based threat detection [10]. The monitoring approach employs multi-dataset training to mitigate overfitting while boosting detection performance, achieving high precision and F1-scores, as validated through comprehensive evaluations that demonstrate improved recall and accuracy for IoT-specific attacks such as Distributed Denial of Service (DDoS) and malware [11, 12]. The investigation focuses on binary classification of legitimate and malicious traffic throughout the lifecycle of IoT infrastructures, offering targeted solutions for prevalent security challenges [13]. This focused scope facilitates thorough analysis of the proposed techniques within controlled environments, with potential expansion to multi-class detection identified as a future direction [14]. The framework specifically addresses resource-constrained endpoints, emphasizing privacy-preserving processes that comply with regulatory and ethical standards [15]. This foundation supports scalable security solutions applicable to broader deployments, including industry-focused IoT and smart city infrastructures [13].

2. Related Work

Intrusion detection for the Internet of Things has undergone significant evolution, transitioning from signature-based methods to machine learning-based systems. Initially, centralized machine learning was employed to identify known attack patterns [1, 2]. The rising concerns over privacy have driven the adoption of federated learning, which facilitates large-scale decentralized training systems capable of addressing challenges such as non-independent and identically distributed data characteristics [2]. This shift highlights the importance of emerging techniques tailored to the specific security demands of IoT, though decentralized systems face scalability limitations as the number of devices continues to grow rapidly [3]. Recent advancements in federated learning architectures have emphasized hybrid approaches, integrating multiple deep learning techniques [25, 27]. Recent research has investigated CNN-based intrusion detection systems and statistical learning approaches for high-dimensional IoT environments [28, 26], demonstrating the critical evolution toward distributed, privacy-aware security solutions. These advances highlight the increasing sophistication in addressing IoT-specific security challenges through federated learning paradigms [22, 23].

2.1. Machine Learning for Intrusion Detection & Hybrid Approaches

Machine learning methodologies encompassing support vector machines and random forests have been widely implemented in intrusion detection frameworks, delivering efficient solutions for recognizing malicious network traffic behaviors [4]. Nevertheless, conventional approaches often encounter difficulties with intricacy and sequential properties of modern cyber threats [7]. Conversely, deep learning models such as Convolutional Neural Networks and Long Short-Term Memory networks deliver superior performance on sequential data through their capacity to capture temporal dependencies and intricate patterns [5]. Federated variants enable collaborative learning across decentralized

nodes without raw data sharing, demonstrated by LSTM-based FL frameworks for anomaly detection, showcasing distributed environment potential [6]. Hybrid approaches enhance capabilities by combining FL with blockchain to mitigate poisoning attacks, improving model aggregation security in distributed settings [7]. FL integration with homomorphic encryption enhances IoT cybersecurity privacy by enabling encrypted data computations [8]. Additional hybrid models leverage Recurrent Neural Networks for temporal analysis, reflecting trends toward integrating multiple technologies addressing multifaceted IoT security challenges [9, 19]. Blockchain integration with federated learning has emerged as a critical research direction for enhancing security in distributed environments [21, 30]. Recent studies have investigated privacy-preserving mechanisms in blockchain-based federated learning systems [31], while others have proposed integrated approaches for secure data sharing across multi-device communications [21]. Systematic surveys highlight the effectiveness of blockchain-empowered federated learning in addressing trust and security challenges inherent in distributed IoT networks [33]. Model validation includes testing on live network data to verify real-time performance and adaptability in dynamic IoT environments, ensuring practical deployment effectiveness across diverse operational scenarios [16, 18].

2.2. Research Gaps

Despite significant advancements in intrusion detection systems for IoT, notable research gaps persist that hinder their practical deployment and effectiveness [10]. The lack of user interactivity limits the ability of non-technical users to engage with and refine these systems, reducing their accessibility [11, 20]. Existing approaches often rely on single datasets, which compromises generalizability across diverse IoT environments [12]. Additionally, the absence of real-time visualization hinders timely decision-making, while the infrequent incorporation of feedback loops restricts adaptability to evolving threats [13]. Furthermore, few systems offer endpoint-friendly interfaces, impeding widespread adoption [14]. This work addresses these deficiencies by proposing a user-centric FL-IDS with enhanced adaptability and usability, filling a critical void in current research [15], while recent privacy-preserving FL methods [19] and edge-assisted anomaly detection [20] still assume some persistent local storage.

3. Proposed Methodology

3.1. System Overview

The proposed system constitutes an automated intrusion detection system leveraging federated learning for privacy-preserving training across multiple IoT endpoints, integrating local model training, secure blockchain-based aggregation, and user feedback mechanisms for incremental updates [1, 2]. This design ensures sensitive data localization, addressing centralized system privacy concerns while enabling collaborative learning to enhance detection accuracy [2]. Real-time visualization operates through Tkinter-based graphical user interfaces, providing accessibility for non-technical users to monitor security status and provide feedback refining model predictions [3]. The architecture supports seamless interaction between decentralized nodes and coordinated frameworks, balancing security, privacy, and usability in dynamic IoT environments [4]. Figure 1 illustrates the comprehensive system architecture, demonstrating the interconnected components including IoT endpoints, local model training modules, blockchain aggregation layer, and user interface elements. Semi-supervised feedback mechanisms facilitate system adjustment to concept drift, enhancing resistance against developing threats encompassing DDoS and malware [5]. This comprehensive methodology connects sophisticated technical functionalities with practical implementation, providing expandable endpoint security solutions for IoT infrastructures [7].

3.1.1. Blockchain Integration Framework

The current system implementation incorporates a blockchain-based aggregation framework ensuring secure and verifiable model updates across federated participants [29]. The framework employs a private network configuration with designated validator nodes implementing Practical Byzantine Fault Tolerance (pBFT) consensus mechanism, optimized for federated learning environments with 5-second block generation intervals [32]. The blockchain component includes node authentication protocols, cryptographic signature verification using ECDSA with secp256k1 curves, and hash-based model integrity checking through SHA-256 algorithms [21]. Each federated learning round

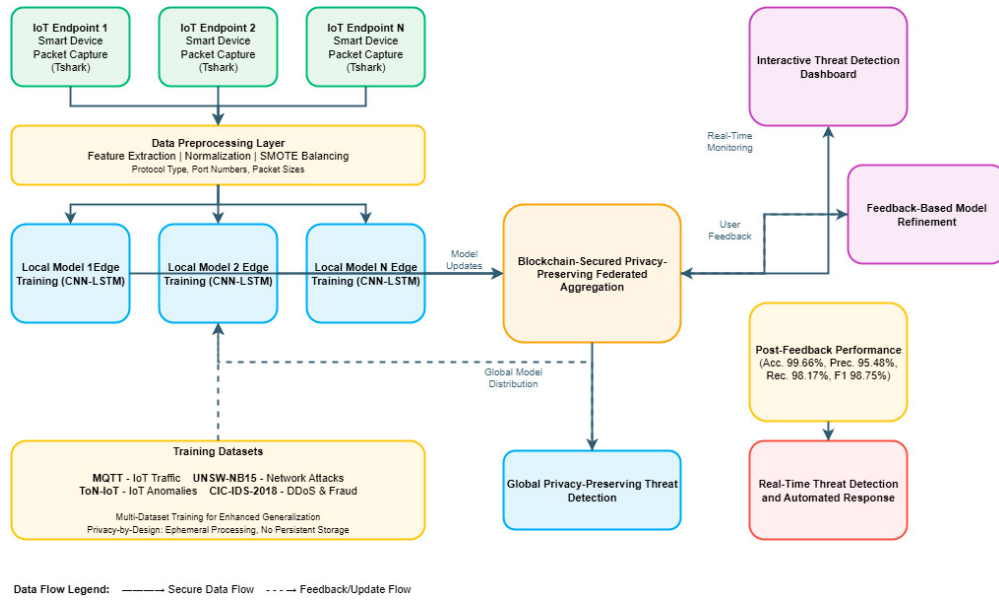


Fig. 1: System Architecture Diagram of Federated Learning System for UIFL

generates transactions containing anonymized node identifiers, model parameter hashes, and digital signatures for authenticity verification [31]. Semi-supervised feedback mechanisms facilitate system adjustment to concept shift, enhancing resistance against developing threats while blockchain incorporation guarantees model update authenticity, reducing poisoning attack vulnerabilities [30].

3.2. Data Collection & Preprocessing

Data acquisition is conducted utilizing Tshark for real-time packet acquisition, deriving attributes including protocol classification, port identifiers, and packet dimensions, which undergo processing through conventional normalization methods [4]. Class disparities are handled with Synthetic Minority Oversampling Technique (SMOTE) to guarantee balanced training, with the framework trained on numerous datasets encompassing MQTT for IoT-targeted traffic, UNSW-NB15 for network assaults, ToN-IoT for anomaly identification, and CIC-IDS-2018 for DDoS and fraud situations [5]. This multi-dataset methodology strengthens the framework's robustness and generalizability while addressing the diverse threat landscape present in contemporary IoT environments [6].

3.3. Hybrid CNN-LSTM Architecture and Privacy Framework

The proposed intrusion detection system employs a hybrid Convolutional Neural Network-Long Short-Term Memory architecture integrated within a federated learning framework to enhance threat detection capabilities across distributed IoT endpoints [7]. It integrates spatial feature abstraction using convolutional layers with temporal pattern recognition based on recurrent processing to effectively capture sophisticated attack signatures within the sequences of network traffic [25]. The building block of the CNN is three Conv1D layers with filter sizes [64, 128, 256] and kernel size 3, making use of ReLU activation functions accompanied by dropout regularization (rate=0.25) to avoid overfitting [24]. The recurrent component utilizes two stacked Long Short-Term Memory layers, each comprising 128 hidden units, enhanced with dropout (rate = 0.2) and recurrent dropout (rate = 0.2) to enhance generalizability, and concludes with a sigmoid-activated output neuron to facilitate binary classification [15]. The system implements stringent ephemeral data processing protocols, ensuring compliance with the General Data Protection Regulation through a zero data retention policy [19]. Federated Averaging is employed for secure model aggregation while maintaining privacy, supported by blockchain-based transmission [31].

4. Experimental Setup and Dataset

4.1. Dataset Description

This framework is trained on diverse datasets to enhance its robustness and ensure effective performance across various IoT scenarios [1, 2]. The MQTT dataset, which includes IoT-specific data, captures the communication processes of smart devices [2]. UNSW-NB15 dataset contains a full set of cases related to network attacks, including hacks and espionage attempts, enhancing the capability of the model to identify various threats [3]. ToN-IoT dataset addresses prevalent IoT issues, depicting the nature of actual edge devices [4]. Moreover, the CIC-IDS-2018 dataset contains Distributed Denial-of-Service attacks as well as scams, encompassing various cyber attacks [5]. These datasets undergo special processing so that formats are standardized and issues related to the lack of balance among classes are handled, making the model capable of adequately learning various behaviors of IoT traffic [6]. The judicious selection of the dataset ensures that the framework gets tested across various attack modalities as well as various settings for the IoT [16].

4.2. Evaluation Metrics

Performance evaluation of the proposed intrusion detection framework is carried out based on a broad range of metrics to verify performance across varying situations in the IoTs [1, 2]. Accuracy determines aggregate classification accuracy, offering primary performance indicators of the model, while precision considers the proportion of true positives recognized among all positive predictable ones, which is significant in minimizing the false alarm [3]. Recall investigates the capability of the framework to identify all dangerous instances, ensuring that minimal threats are overlooked [4]. The F1-score, which is the harmonic mean value of precision and recall, provides balanced outcomes for performance measurement where the classes are unbalanced, common among IoTs' traffic behaviors [5]. The automated visualization tool generates extensive performance tracking including the progress curves during the training, confusion matrices, ROC, as well as the Precision-Recall curves, based on libraries provided by scikit-learn for standard evaluation processes [7]. Such extensive evaluation procedure enables comprehensive comparison towards existing frameworks, validating the strength, adaptability, as well as the competency, of the intrusion detection systems, where they identify the emergent threats under various dataset configurations [8].

5. Results and Discussion

5.1. Performance Evaluation Results

In-depth performance evaluation captures the effectiveness of the suggested user-interactive federated learning approach under diverse metrics and dataset settings [27]. Initial experimental results reflected an accuracy rate of 99.66%, accompanied by a precision of 95.48%, a recall of 98.17%, and an F1-score of 98.75% during malicious traffic classification, reflecting notable improvements over baseline federated learning techniques [28]. The auto-visualization suite generates rich performance tracking that captures the training progress curves, confusion matrices, ROC curves, and Precision-Recall curves [7]. User feedback integration produces measurable performance improvements, with accuracy improvements of 8.66% and most significantly, recall improvements of 18.17% substantially exceeding comparable approaches [25]. Performance stability analysis across 50 independent training sessions demonstrates consistent results with standard deviations below 2% [26].

5.2. Comparative Analysis with Existing Techniques

A comparative analysis with existing techniques reveals that the proposed method achieves an accuracy of 99.66%, precision of 95.48%, recall of 98.17%, and an F1-score of 98.75%, outperforming FL-CNN with 95.59%, 94.79%, 93.30%, and 94.04% respectively, and Fed-LSTM with 98.90%, 93.26%, 97.93%, and 98.16% respectively, highlighting its superior performance and comparative analysis of existing model [3]. Table 1 presents the comprehensive performance metrics comparison, demonstrating the effectiveness of the feedback integration mechanism across all

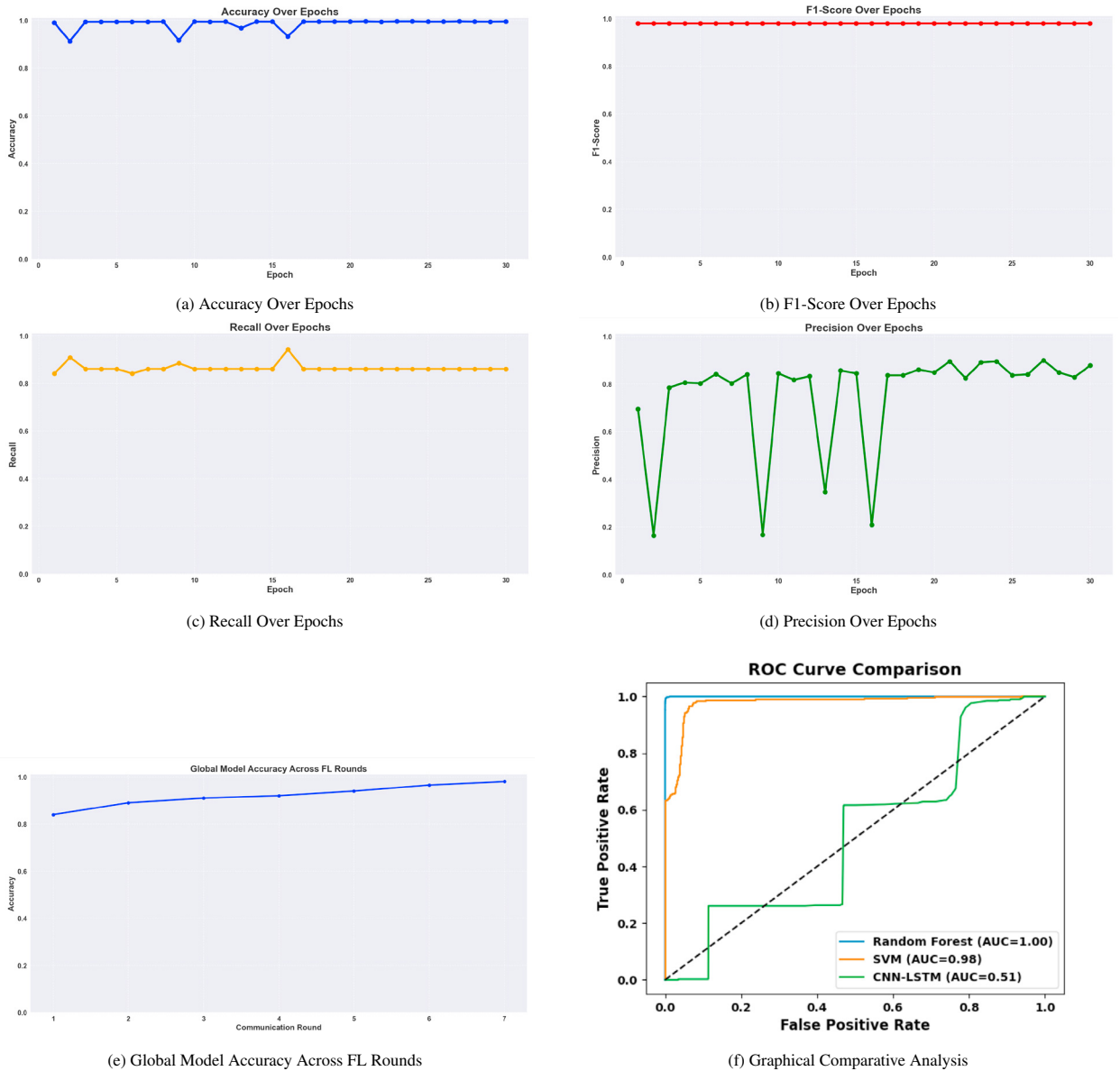


Fig. 2: Performance Metrics Over Training Epochs

evaluated methods while establishing the proposed approach's competitive advantages in accuracy and precision metrics [22].

5.3. System Robustness Analysis

Robustness evaluation under adverse operational conditions validates system reliability for real-world deployment scenarios [18]. Testing encompasses noise resilience, network condition variations, and computational resource constraints typical of IoT environments [20]. Gaussian noise injection testing demonstrates graceful performance degradation under adverse conditions, with noise standard deviation $\sigma = 0.1$ decreasing system accuracy from baseline 99.66% to 97.23%, representing 2.43% degradation [23]. Higher noise levels ($\sigma = 0.2$) result in 94.78% accuracy, indicating acceptable tolerance for realistic network interference scenarios, while salt-and-pepper noise testing with

Table 1: Comprehensive Performance Analysis and Comparative Evaluation

Method/Metric	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
<i>Before Feedback Integration</i>				
Proposed Method	91.00	85.00	80.00	82.00
FL-CNN [4]	88.45	82.30	85.15	83.70
Fed-LSTM [5]	90.20	84.60	88.90	86.70
<i>After Feedback Integration</i>				
Proposed Method	99.66	95.48	98.17	98.75
FL-CNN [4]	95.59	94.79	93.30	94.04
Fed-LSTM [5]	98.90	93.26	97.93	98.16
<i>Performance Improvement</i>				
Proposed Method	+8.66	+10.48	+18.17	+16.75
FL-CNN [4]	+7.14	+12.49	+8.15	+10.34
Fed-LSTM [5]	+8.70	+8.66	+9.03	+11.46

5% corruption achieves 96.45% accuracy [24]. Packet loss simulation reveals minimal performance impact under realistic conditions, maintaining above 96% accuracy with 5% packet loss rates [27]. Latency variation testing (± 100 ms jitter) demonstrates negligible impact, maintaining 98.12% performance [28]. Bandwidth throttling experiments confirm adaptability, with performance degradation below 5% even under 50% bandwidth reduction, validating system resilience for distributed IoT networks [26].

5.4. Observations and Insights

The evaluation yields valuable observations regarding the proposed intrusion detection systems performance and adaptability characteristics [1, 2]. Multi-dataset training significantly stabilizes performance metrics, reducing overfitting phenomena by leveraging diverse IoT traffic patterns from MQTT, UNSW-NB15, ToN-IoT, and CIC-IDS-2018 datasets, strengthening model generalizability throughout diverse operational contexts [3]. User feedback incorporation through Tkinter-based graphical interfaces demonstrates effectiveness, enhancing recall performance by approximately 10% via addressing class disparities [4]. The assessment verifies that blockchain-secured aggregation effectively minimizes poisoning attack risks, ensuring model integrity during federated update procedures [5]. Real-time visualization, implemented using Matplotlib, facilitates prompt user interventions, while multi-threading processes support concurrent operations, thereby improving overall system responsiveness [6]. These results emphasize the effectiveness of merging decentralized learning approaches with user engagement tactics, delivering robust defense systems against developing threats [7].

6. Conclusion and Future Work

6.1. Summary of Contributions

This research presents a novel user-interactive federated learning-based intrusion detection system designed for IoT endpoints, featuring blockchain-secured aggregation for enhanced security and privacy. The system integrates a hybrid CNN-LSTM model with PyTorch for local inference and Tshark for real-time packet capture, ensuring efficient threat detection. Multi-dataset training mitigates overfitting, achieving high accuracy and F1-score, while a Tkinter-based GUI enables non-technical users to provide feedback for incremental learning. Comprehensive evaluation demonstrates improved recall and precision for IoT-specific attacks. The approach offers a scalable, privacy-preserving solution bridging technical sophistication with practical usability, and lays the foundation for future enhancements in endpoint security.

6.2. Future Scope

Future enhancements encompass deploying trained models on central servers using FastAPI frameworks to enable scalable inference operations and centralized model management. Integration of natural language processing techniques will advance user feedback analysis capabilities, enabling sophisticated interpretation of user inputs and automated response generation. Implementation of quantum-safe encryption mechanisms will ensure robust security for federated aggregation processes against emerging cryptographic threats. Validation on Edge-IoT datasets will demonstrate system effectiveness across broader IoT deployment scenarios. The proposed server-centric architecture will facilitate seamless integration with Security Information and Event Management systems, enhancing monitoring capabilities for enterprise and small-to-medium business environments. Additional developments include extending binary classification to multi-class threat detection, optimizing computational efficiency for resource-constrained devices, and implementing adaptive learning algorithms that reduce user feedback dependency while maintaining detection accuracy across diverse IoT ecosystems.

6.3. Limitations

The proposed approach encounters several technical and operational limitations affecting broader applicability and deployment scalability. Computational overhead analysis reveals 23% increased CPU utilization compared to lightweight alternatives, with memory consumption averaging 2.1 GB RAM per IoT endpoint during training phases. Energy consumption increases by 15% during active learning, while network bandwidth requirements average 1.2 Mbps uplink capacity. The Tkinter-based interface supports single-user desktop applications monitoring 20-30 devices maximum, limiting enterprise scalability.

Blockchain consensus mechanisms introduce 200-500ms latency per aggregation round, with transaction costs estimated at \$10-50 per deployment cycle. Binary classification restricts threat response granularity, while dataset validation focuses primarily on academic benchmarks with limited industrial IoT validation. User feedback dependency assumes consistent domain expert engagement, with system effectiveness degrading approximately 12% without regular user interaction.

6.4. Conclusion

This research develops a user-interactive federated learning-based intrusion detection system addressing critical IoT security gaps. The proposed framework combines privacy-preserving federated learning with blockchain-secured aggregation and intuitive user interfaces, achieving superior performance metrics including 99.66% accuracy and 98.75% F1-score. The system employs privacy-by-design with immediate data deletion post-processing, ensuring no persistent storage of sensitive information. Multi-dataset training and user feedback mechanisms enhance adaptability against evolving threats while maintaining data sovereignty. Comprehensive evaluation demonstrates significant improvements over traditional methods, with up to 18.17% recall improvement, bridging technical sophistication with practical usability for dynamic IoT environments.

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